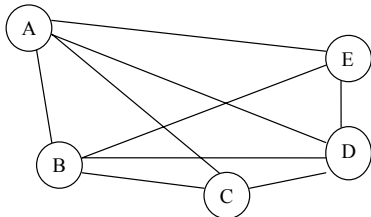


Algorithmic Search

- Exhaustive search for an element with a given property (searching for a needle in a haystack).
- It is the last resort in addressing computational tasks where there is a quick way of verifying that a given element is the answer once you have it, but no *easy* way of constructing the answer.
- Example: finding an Hamiltonian cycle in a graph is classically a *difficult* task (NP-complete).



Grover's Algorithm

- On a conventional classical computer, searching an **unstructured** space of N items takes $O(N)$ steps.
- Quantum computers can do far better. They can solve the search problem in $O(\sqrt{N})$.
- The algorithm for quantum search is due to Lov Grover
- Grover's algorithm is optimal: no search algorithm with faster than $\sqrt{N}$ scaling is possible. This was proved in the 1997 paper.
- The quantum search algorithm still provides a silver bullet for speeding up a wide class of classical computations.
- From a practical viewpoint, this quadratic speed-up makes a big difference.

Grover, Lov K. "A framework for fast quantum mechanical algorithms", 1997

Practical Impact of Grover's algorithm

Suppose $N = 10^{30}$.

- With a quantum processor capable of 10^8 calls/s we could find a solution with 10^{15} calls, i.e. in 10^7 seconds (about 4 months).
- With a classical processor of the same speed we would need about 10^{22} seconds, or very nearly the age of the universe.

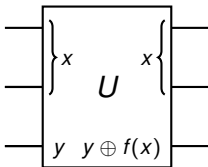
Impact on **cryptography**: need to increase substantially the length of cryptographic keys in order to survive *brute force attacks*.

Cryptoanalysis remains nevertheless a **difficult** problem (NP-complete).

Oracle Computation

Unitary evolution of quantum systems implies **reversibility**.
Thus classical irreversible functions must first be made reversible.

$$f : \{0, 1\}^n \rightarrow \{0, 1\} \text{ becomes } U_f : |x\rangle |y\rangle \mapsto |x\rangle |y \oplus f(x)\rangle$$



In quantum algorithms U_f is treated as a *black box* or *oracle*.
Complexity is given by the number of **queries** (= oracle invocations).

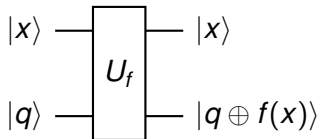
Setting the Problem

Consider a search space of $N = 2^n$ elements, to which we'll refer to by indices $0, 1, \dots, N$.

Define the function $f : \{0, 1\}^n \rightarrow \{0, 1\}$

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is not a solution} \\ 1 & \text{if } x \text{ is a solution} \end{cases}$$

Define the quantum oracle



Note that if we initialise q to 0, then it is flipped if $f(x) = 1$.

Grover's Iteration

The main component of the quantum search algorithm is the **Grover operator**

$$G = H^{\otimes n} P_0 H^{\otimes n} O$$

The oracle O is as above with $q = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$. This allows for the ancilla qubit to stay unchanged during the iteration:

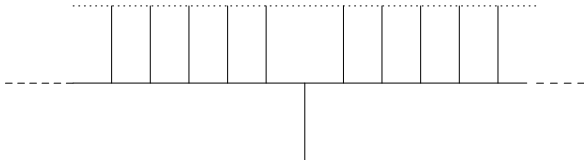
$$|x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \xrightarrow{O} (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The operator $H^{\otimes n} P_0 H^{\otimes n}$ is called *reflection about the mean* and has the effect of increasing the amplitude of solutions twice above the mean.

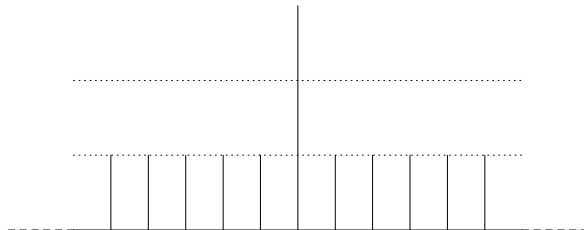
$$P_0 |x\rangle = \begin{cases} |x\rangle & \text{if } x = 0 \\ -|x\rangle & \text{otherwise} \end{cases}$$

Amplitude Amplification

■ Effect of the oracle O

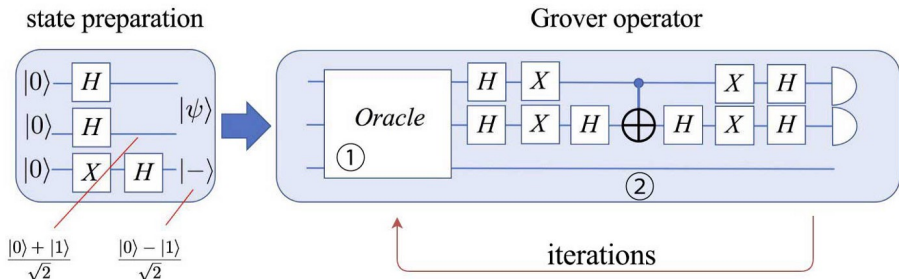


■ Inversion about the mean



Search Procedure

Grover's search consists in iterating the operator G , $O(\sqrt{N})$ times and then measuring the input register.



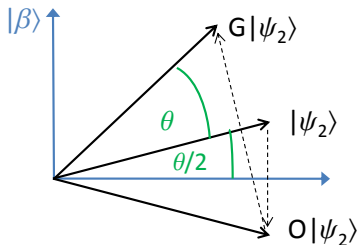
- 1 flip the amplitude if $f(x) = 1$
- 2 flips every amplitude around the average.

We repeat this process $O(\sqrt{N})$ times to amplify the amplitude of the right answer.

Geometrical Description

Consider a plane (β, α) with $\beta = t$, the only solution to the search problem. The uniform superposition of all candidates is

$$|\psi_2\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle = \sqrt{\frac{N-1}{N}} \sum_{x \neq t} |x\rangle + \sqrt{\frac{1}{N}} |t\rangle$$



$\sin \theta/2$ represents the initial probability of success.

Complexity of Grover's search

How many queries?

Geometrically: how many rotations?

We need to hit the target t . Thus

$$\frac{\vartheta}{2} + n\vartheta = \frac{\pi}{2},$$

where $\frac{\vartheta}{2} = \arcsin(\frac{1}{\sqrt{N}})$.

Since for small x , $x \sim \arcsin x$, we can approximate n as follows:

$$n \sim \frac{\pi}{2\vartheta} \sim \frac{\pi}{4 \arcsin(1/\sqrt{N})} \sim \frac{\pi}{4} \sqrt{N}.$$